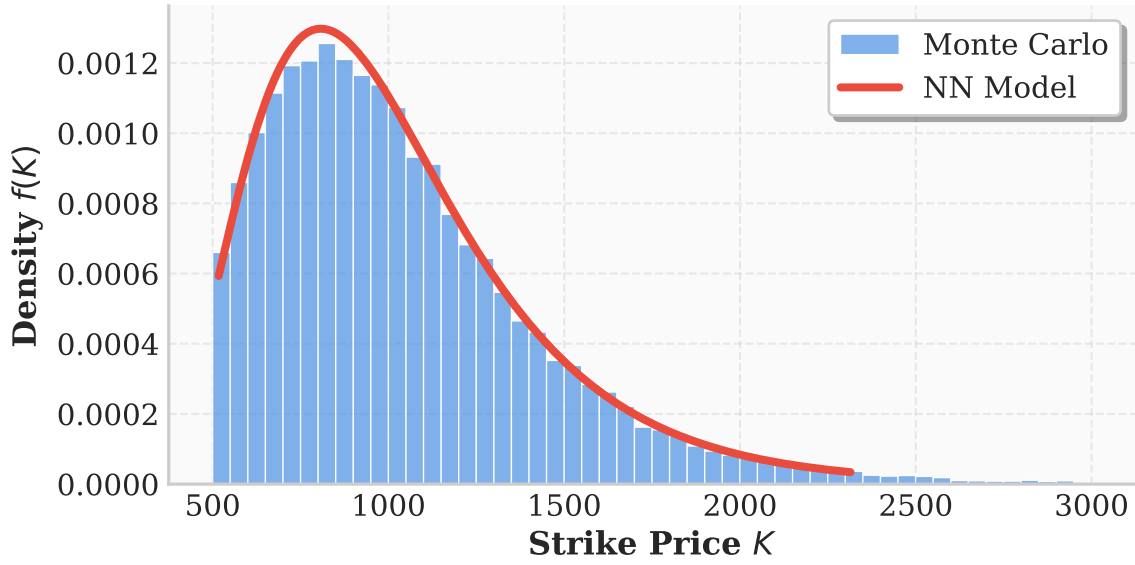


PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S)S_t dW_t$

Pretrained (corrected phi_tilde mapping)

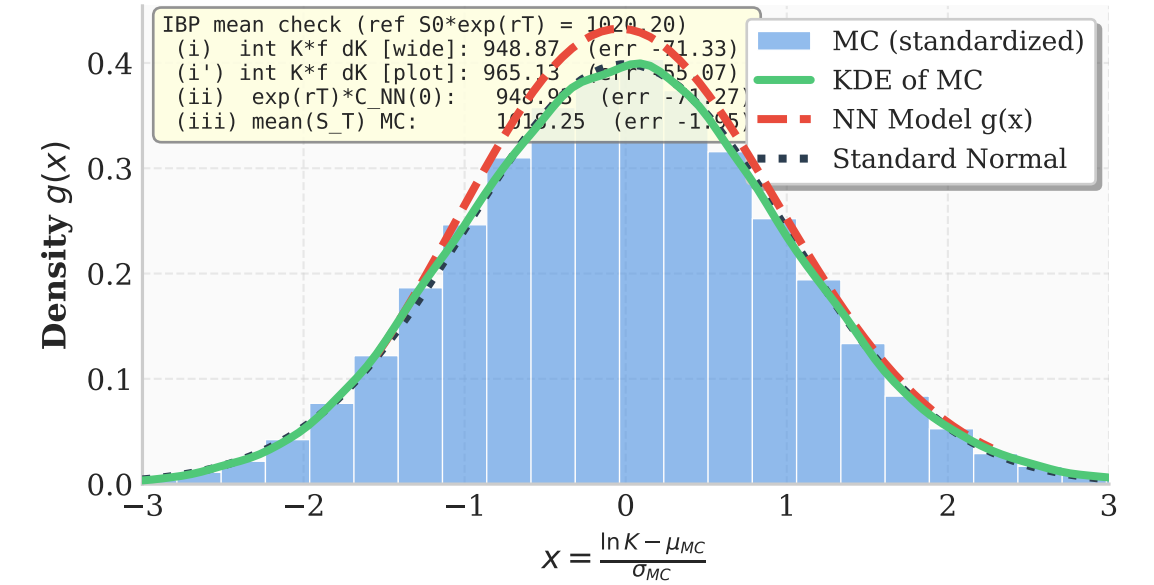
Strike Distribution (T = 0.50)



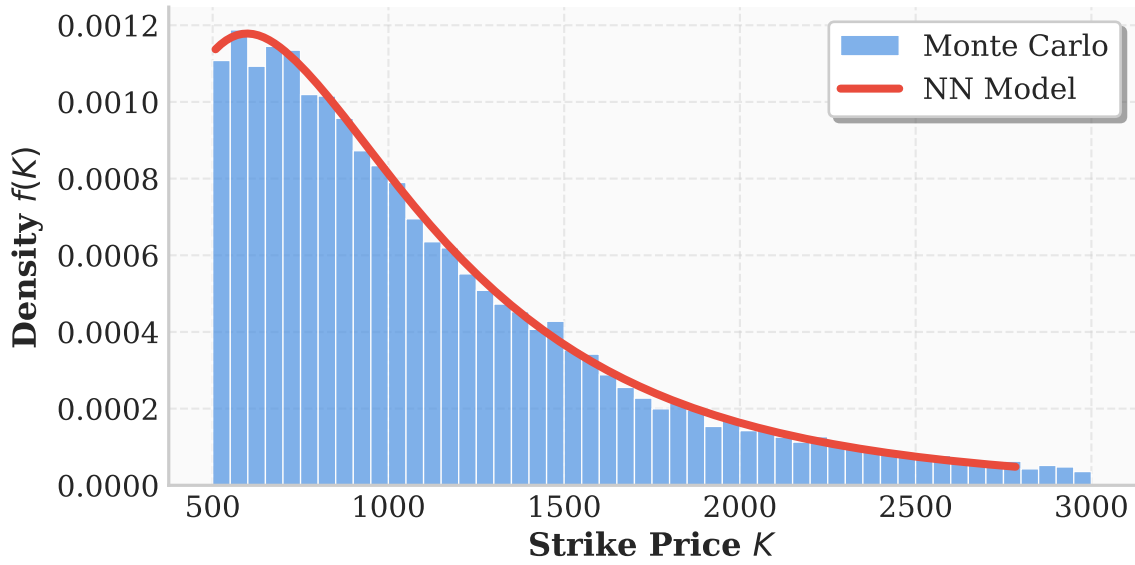
Log-Normal Distribution (T = 0.50)



Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.85, \sigma_{MC}=0.38 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.88, \sigma_{model}=0.33$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.06, \text{ExKurt}=0.04$



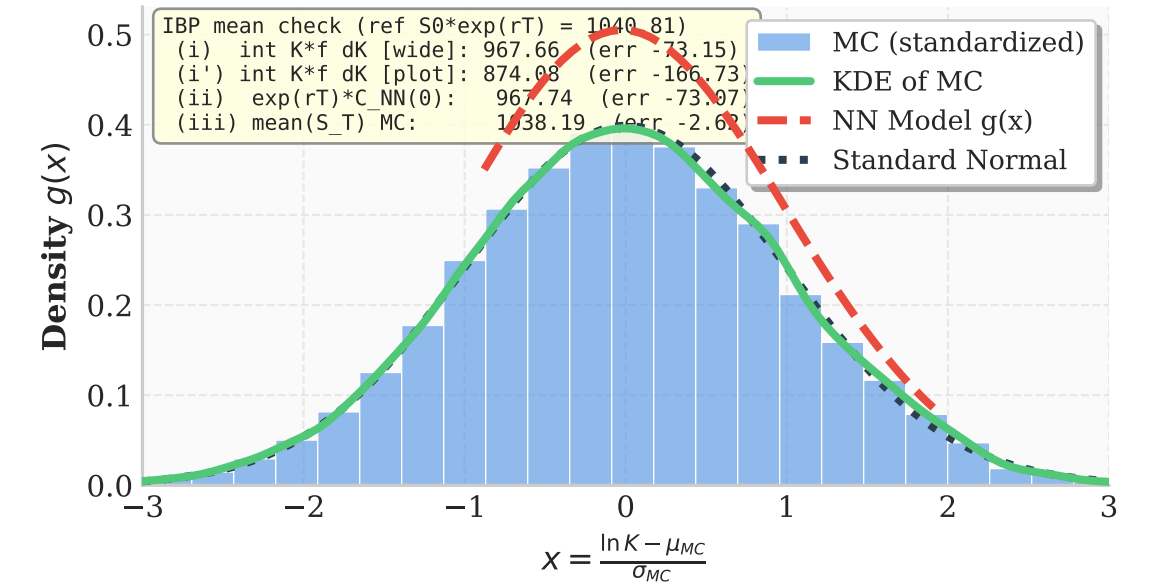
Strike Distribution (T = 1.00)



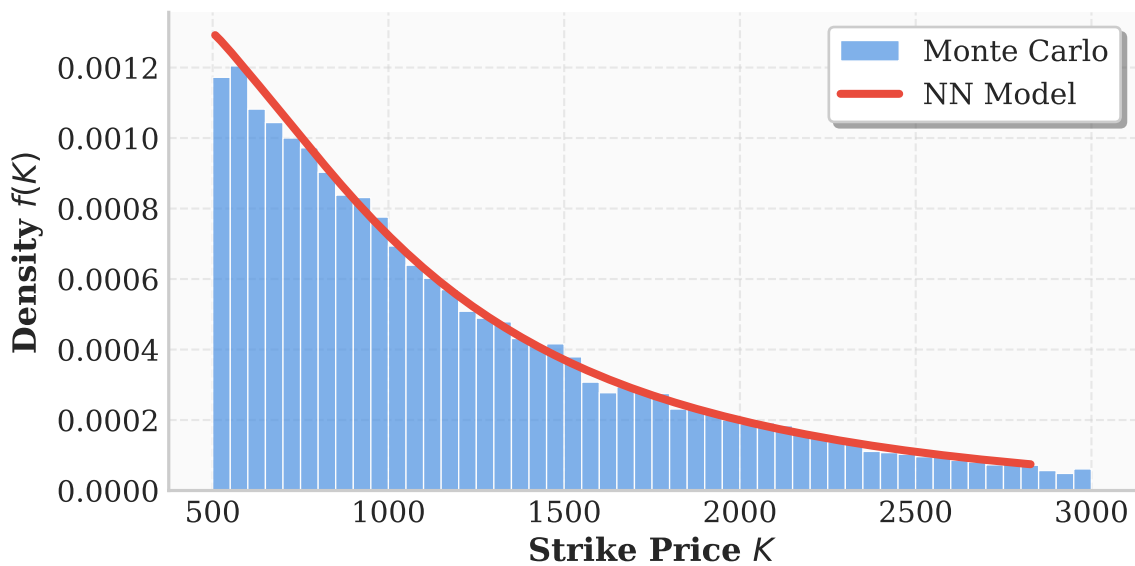
Log-Normal Distribution (T = 1.00)



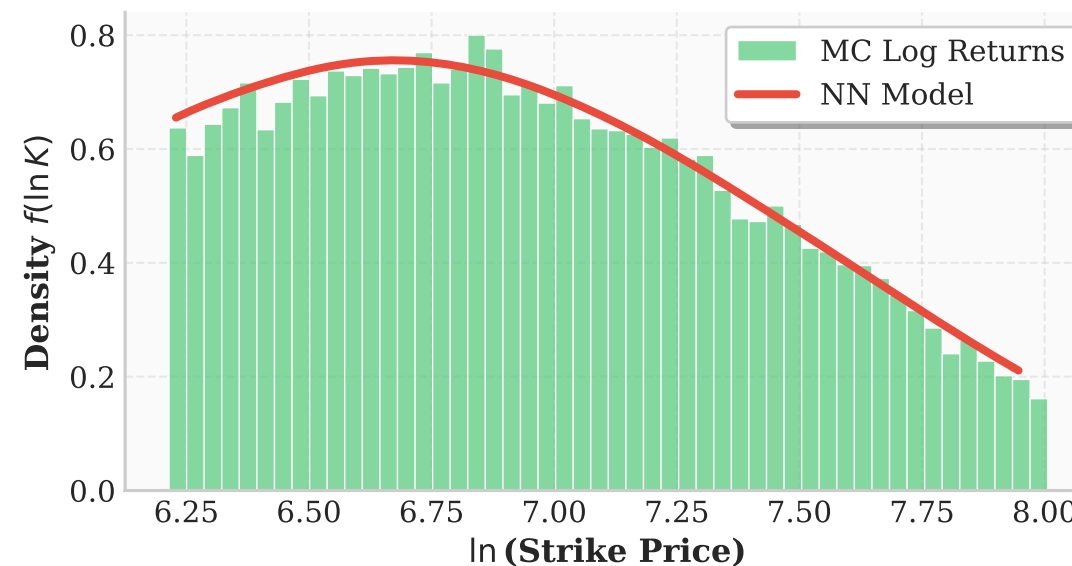
Gaussian Space (T = 1.00)
MC Standardization: $\mu_{MC}=6.77, \sigma_{MC}=0.60 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.92, \sigma_{model}=0.43$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.01, \text{ExKurt}=-0.06$



Strike Distribution (T = 1.50)



Log-Normal Distribution (T = 1.50)



Gaussian Space (T = 1.50)
MC Standardization: $\mu_{MC}=6.68, \sigma_{MC}=0.75 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.96, \sigma_{model}=0.45$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.07, \text{ExKurt}=-0.14$

